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Unit – 4 : Session – 11 : SLO – 2 | Converting a TCP Client to UDP

■ I. Key Changes (TCP client → UDP client)

1. Socket type: `socket.SOCK_STREAM` → `socket.SOCK_DGRAM`.
2. No connection setup: remove `connect()` — UDP is connectionless.
3. Sending: `send()/sendall()` → `sendto(data, (host, port))` — every datagram carries its destination.
4. Receiving: `recv()` → `recvfrom(bufsize)` — returns (data, address) because the sender must be identified.
5. No `listen()/accept()` on the peer; a UDP "server" just `bind()`s and calls `recvfrom()`.
6. Error handling: no automatic retransmission — timeouts and retries must be implemented by the application.

■ II. Example — UDP Client

- `import socket`
- `s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)`
- `s.sendto(b"Hello", ("localhost", 5000))`
- `data, addr = s.recvfrom(1024)`
- `print(data); s.close()`